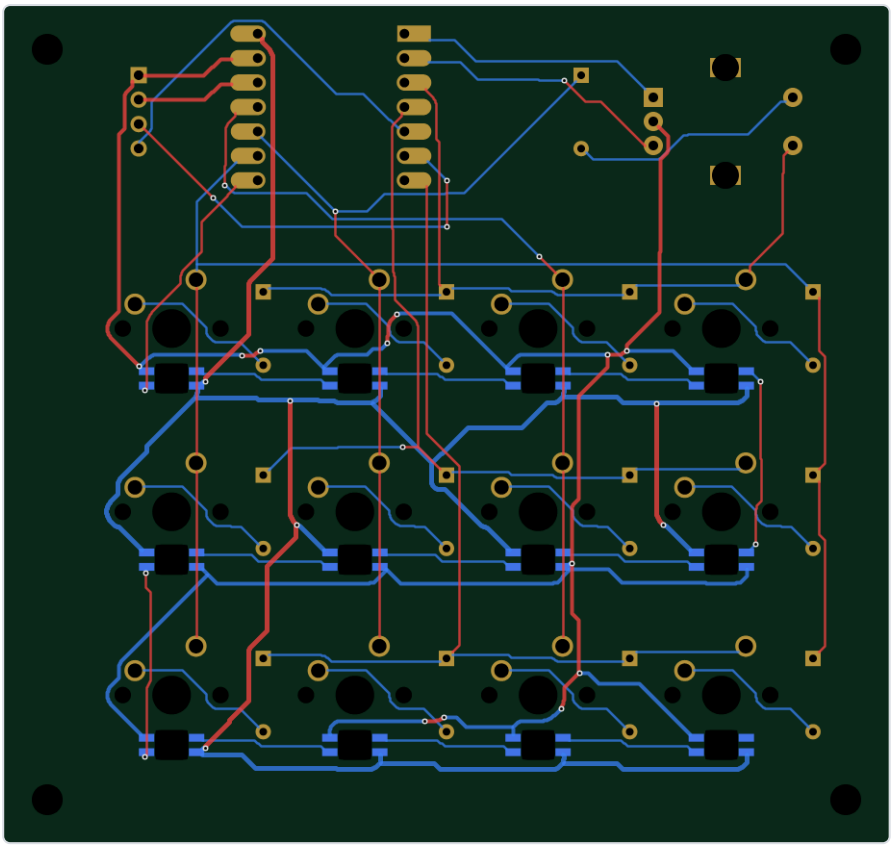
TechPad PCB Reference

The finished board as of 18 Oct 2026: 0 DRC errors, ground fill on both layers. This replaces the earlier placement and routing notes.

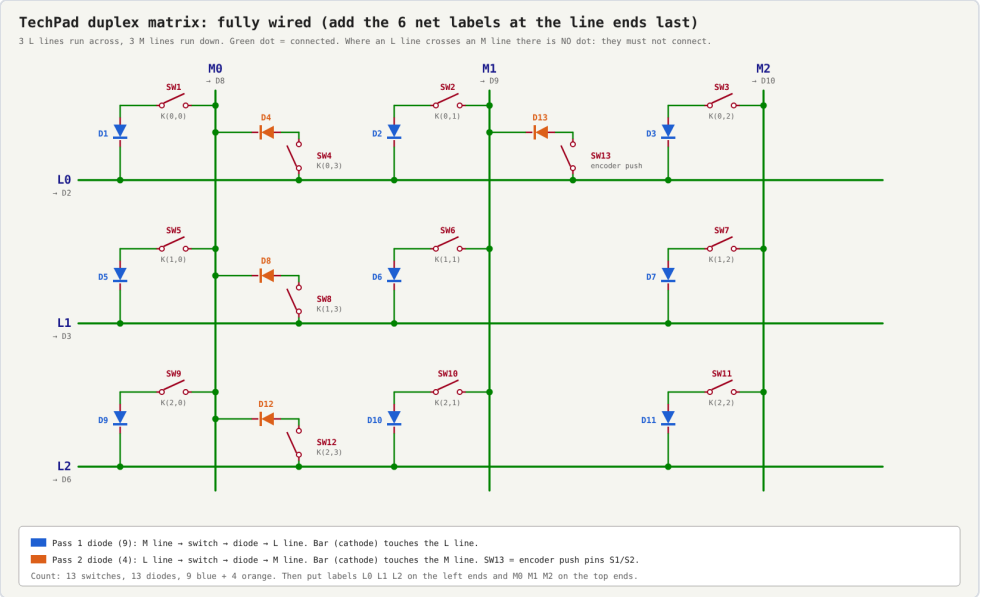


Routing: red = front copper (F.Cu), blue = back copper (B.Cu). 752 track segments, 31 vias, plus the GND fill.

1. Pin map (XIAO RP2040)

XIAO pin	Net	Goes to
D0	ENC_A	Encoder A
D1	ENC_B	Encoder B
D2 / D3 / D6	L0 / L1 / L2	Duplex matrix "L" lines
D4 / D5	SDA / SCL	OLED I²C
D7	DIN	LED1 data in (the chain then runs LED1 → LED12)
D8 / D9 / D10	M0 / M1 / M2	Duplex matrix "M" lines
5V	+5V	All 12 LEDs
3V3	+3V3	OLED VCC
GND	GND	Everything, plus the copper fill

Seen from the front, the XIAO's pins are mirrored, because it sits on the back. Left column, top to bottom: 5V, GND, 3V3, D10, D9, D8, D7. Right column: D0 to D6. Certain



Duplex matrix: each pair of lines reads two keys, one per diode direction.

2. Part positions

KiCad page coordinates, and the same point in case coordinates (origin at the PCB's bottom-left corner, Y up).

Part	KiCad (mm)	Case X/Y (mm)
SW1 to SW4 (top row)	centres (117.425 / 136.475 / 155.525 / 174.575, 133.525)	(17.425, 36.475, 55.525, 74.575; 53.475)
SW5 to SW8 (middle row)	same X, Y 152.575	Y 34.425
SW9 to SW12 (bottom row)	same X, Y 171.625	Y 15.375
D1 to D12 (1N4148, THT)	X 126.95 / 146.0 / 165.05 / 184.1, Y 129.715 / 148.765 / 167.815, rotation −90	right of each switch
D13 (knob click diode)	(160.0, 107.2), rotation −90	(60.0, 79.8)
LED1 to LED12 (SK6812 MINI-E)	switch centre + 5.08 down; rows 1 and 3 rotation 180, row 2 rotation 0	snake: row 1 L→R, row 2 R→L, row 3 L→R
SW13 encoder EC11	anchor (167.5, 109.5), shaft (175.0, 112.0)	(75.0, 75.0)
J1 OLED header 1×4	pin 1 GND (114.0, 107.19), pins every 2.54 down	(14.0, 79.81) … (14.0, 72.19)
U1 XIAO RP2040	(134.0, 110.5), rotation 180, **back side**	centre (34.0, 76.5), USB-C at top edge
H1 to H4 M3 holes	(104.5,104.5) (187.5,104.5) (104.5,182.5) (187.5,182.5)	(4.5,82.5) (87.5,82.5) (4.5,4.5) (87.5,4.5)
Board outline	(100,100) to (192,187)	(0,0) to (92,87)

3. Design rules in use

Rule	Value

Track width	Signals 0.25, GND 0.4, +3V3 0.4, +5V 0.5
Clearance	0.25 (the XIAO pads require 0.25)
Vias	0.6 pad / 0.3 drill
Copper to board edge	0.3 (the router kept 0.5)
Custom rule pcb.kicad_dru	Lets the LED pads sit close to their own footprint cut-outs. The real board edge still keeps the 0.3 rule
GND fill	F.Cu and B.Cu, clearance 0.3, min width 0.25, thermal reliefs

4. DRC status and the warnings that are left

- **0 errors and 0 unconnected pads.** Certain (KiCad DRC, 18 Oct)
- About 19 yellow silkscreen warnings remain, from the XIAO footprint's own outline. It crosses its pads and the top edge, where the USB-C hangs over. That outline comes with the footprint, so these are harmless.
- Reference labels for D, LED and H parts are hidden on purpose. The parts are identical, and the labels overlapped the switches.
- The back silkscreen reads "TechPad v1.0 / by MoonRock".

5. Backups on your PC (B:\Aarav\My_TechPad\pcb)

File	What it is
pcb-before-placement.kicad_pcb.bak	Before the parts were placed
pcb-before-routing.kicad_pcb.bak	Your partial routing, before the auto-route
pcb-before-reroute.kicad_pcb.bak	The first routing, which had the XIAO pad errors
pcb-before-silk.kicad_pcb.bak	Routed, before the silkscreen tidy

6. Exports for manufacturing Likely

> **Gerbers.** File → Fabrication Outputs → Gerbers. Output folder `gerbers/`. Tick these layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts. Click **Plot**.

> **Drill files.** In the same dialog, click **Generate Drill Files...**. Excellon format, same folder, then Generate.

> **Zip and check.** Zip the folder as **gerbers.zip** and drop it into a free online Gerber viewer. You should see the 92 × 87 outline, both copper layers, the GND fill and your text. A missing Edge.Cuts layer is the most common mistake.

> **STEP.** File → Export → STEP. Origin: User defined, X 100, Y 187. Save as pcb.step. This is for the case work.